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Next item →

1. What is the main use of mathematical induction?

1 / 1 point

☐ Mathematical induction is never useful to assert that a propositional function $P(n)$ is true for all positive integers n

☒ Mathematical induction can be used to prove $P(n)$ is true for all integers greater than a particular integer, where $P(n)$ is a propositional function

☐ Mathematical induction is always more difficult than using a direct proof

✔ Nice work
2. What is the rule of inference behind mathematical induction?

1 / 1 point

☒ $P(1)$ is true, $\forall k \in \mathbb{N} (P(k) \rightarrow P(k + 1))$, therefore $\forall n \in \mathbb{N} P(n)$

☐ $P(1)$ is true, $\forall k \in \mathbb{N} P(1), P(2) \dots P(k) \rightarrow P(k + 1)$, therefore : $\forall n \in \mathbb{N}, P(n)$

☐ Hypothetical syllogism

☐ Universal instantiation

✔ Nice work
3. What are the main steps in building a mathematical induction ?

1 / 1 point

☒ Show that $\forall k \in \mathbb{N}$: if $P(k)$ is true, called inductive hypothesis, then $P(k + 1)$ is true

☐ Show that $P(10)$ is true

☐ Show that $\exists k \in \mathbb{N}$: if $P(k)$ is true, then $P(k + 1)$ is true

✔ Nice work
4. What are the intuitions behind mathematical induction?

1 / 1 point

☐ Every step is completely independent from the previous one

☒ Mathematical induction is like climbing an infinite ladder

☐ The base case shows that the property initially does not hold true

✔ Nice work
5. Which of the following is an example of the use of mathematical induction?

1 / 1 point

☒ Proving formulas

☐ Building all the predicate logic

☐ Proving all the rules of inference

☐ Definition of all the Boolean algebra

✔ Nice work